Universidade de Brasília
Instituto de Ciências Exatas
Departamento de Ciência da Computação

Layout-Aware Zero-Shot Learning for Visual Document Matching

Lucas de Almeida Bandeira Macedo

Dissertação apresentada como requisito parcial para conclusão do Mestrado em Informática

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Resumo

Garantir conformidade documental requer uma identificação acurada dos documentos, que também serve o propósito de manter o dado consistente ao decorrer das esteiras de verificação. Grande parte dos estudos lidam com a identificação como uma tarefa de classificação documental, ou como uma tarefa de segmentação. Entretanto, documentos industriais estão sempre mudando sua forma, e os modelos que os classificam precisam de constantes atualizações. Nesses casos, avaliar se determinado documento está de acordo com o histórico de documentos aceitos é uma abordagem mais apropriada. Esta tese adentra no problema de comparar a aparência de dois (ou mais) documentos para determinar se eles dividem ou não a mesma disposição de informações. Portanto, esse problema é atacado com o paradigma *Zero-Shot Learning* (ZSL), que é uma abordagem poderosa para cenários onde as classes encontradas na inferência não coincidem com as classes usadas no treino. Para dar suporte ao estudo, o *Layout-Aware Complex Document Information Processing* (LA-CDIP) é introduzido, um dataset contendo 4,993 documentos, distribuídos por 144 classes, reorganizadas a partir da base de dados *Ryerson Vision Lab Complex Document Information Processing* (RVL-CDIP), realizando uma separação prioritariamente sintática, ao invés de semântica. Essa abordagem é testada usando redes siamesas e *Contrastive Learning* através de muitas arquiteturas neurais conhecidas, incluindo ResNet, EfficientNet e *Vision Transformer* (ViT). Em cenários ZSL, o método proposto atinge um *Equal Error Rate* (EER) abaixo de 5% na verificação com validação cruzada. Além disso, a abordagem *Visual Document Matching* (VDM) performa com maior precisão que *Large Language Models* (LLMs) de código aberto e rivaliza contra o modelo GPT-4o, da OpenAI, demonstrando a superioridade de uma técnica especialista sobre modelos multimodais generalistas. Essas descobertas mostram que a abordagem proposta mantém alta acurácia enquanto usa significativamente menos parâmetros que LLMs, demonstrando um uso mais prático para aplicações de conformidade documental na indústria.

Palavras-chave: Redes Neurais Artificiais, Análise de Documentos, Aprendizado Tiro-Zero, Tese de Mestrado

Abstract

Ensuring document compliance requires accurate document identification, which plays a crucial role in maintaining consistency throughout document analysis pipelines. Several studies approach layout identification as a document image classification or segmentation task. However, due to the ever-changing nature of industry documents, a traditional classification with entropy learning is often insufficient, as models require frequent retraining. In these cases, determining whether two or more documents share the same visual layout is a more suitable approach. This paper addresses the problem of matching the visual appearance of two (or more) documents to determine whether they share the same layout. To achieve this, a Zero-Shot Learning (ZSL) approach is adopted, which is a powerful technique for scenarios where training classes do not align with those encountered during inference. Layout-Aware Complex Document Information Processing (LA-CDIP) is introduced to support the study, a dataset comprising 4,993 documents across 144 classes, which is reorganized from the Ryerson Vision Lab Complex Document Information Processing (RVL-CDIP) database to emphasize visual structure over semantic content. This approach is benchmarked using a siamese network and contrastive learning framework across multiple backbone architectures, including ResNet, EfficientNet, and Vision Transformer (ViT). In zero-shot scenarios, the proposed method achieves an Equal Error Rate (EER) below 5% in 1-vs-1 verification with cross-validation. Furthermore, the Visual Document Matching (VDM) approach outperforms lighter Large Language Models (LLMs) and rivals GPT-4o, highlighting the superiority of specialized techniques over general-purpose multimodal models. These findings show that the proposed approach maintains high accuracy while using significantly fewer parameters than large multimodal models, making it more practical for real-world document compliance applications.

Keywords: Artificial Neural Networks, Document Analysis, Zero-Shot Learning, Thesis

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Chapter 1

Introduction

1.1 Contextualization

Documents are at the core of corporate society, and they often hold immeasurable value, representing contracts, employment relationships, loans, and identities. Recently, as technology becomes more accessible, documents have become increasingly more digital, some never being printed on paper. These documents often have all important information already separated in the file's metadata and require no extra intelligence to classify or extract the information. However, many organizations still receive documents in physical form, for example, when dealing directly with individual customers, which makes full digitization impractical in many cases.

In such a scenario, the company must have a document understanding pipeline, where the document is approved regarding its form (verifying if the client sent the right type of document) and the information is extracted to support business decision-making, as seen in Figure 1.1. This pipeline has historically been executed by humans. Between the start of the digital revolution and the popularization of Optical Character Recognition (OCR) engines [1, 2], the typist profession was highly prevalent, with workers responsible for manually entering information from physical documents into digital systems. Moreover, the classification and information extraction were done by back office analysts, who spent their workforce reading and verifying if the document met the company's compliance rules. Today, with current technology, especially with the popularization of Neural Networks (NNs), this manual document pipeline has become unthinkable in big companies, as it is inefficient in both time and cost. This work focuses on automatic document classification.

Document Classification is defined by the categorization of documents into predefined classes [3]. Document Image Classification (DIC) is a more specialized task in which textual information is not directly available, and the input often consists of photos or scans of physical documents. Therefore, if the text of the document is ever required,

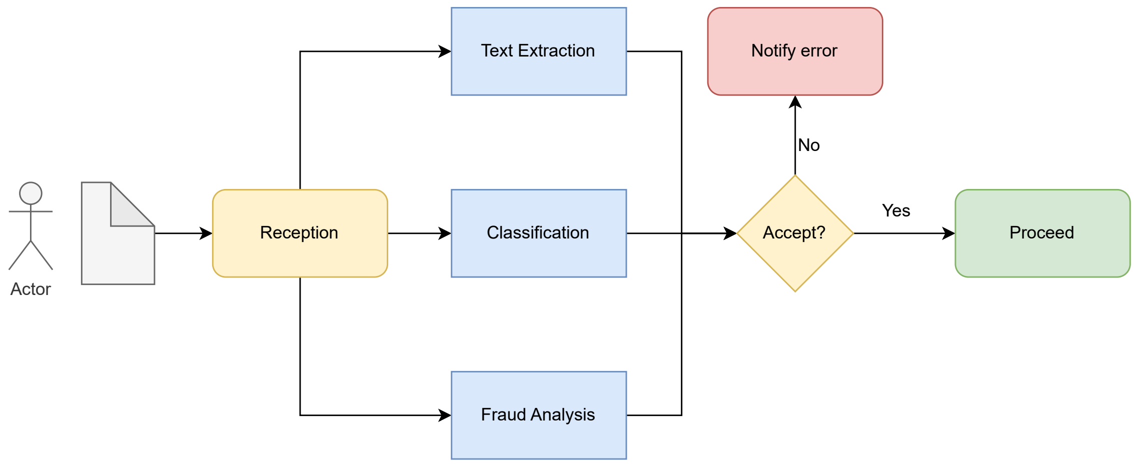

Figure 1.1: Simplified view of a document compliance pipeline. In this diagram, text extraction, document classification and fraud analysis are being used together to decide upon the acceptance of the received document.

this text should be extracted with OCR algorithms. Artificial Intelligence (AI) and Deep Learning (DL) techniques can automate document classification by leveraging traditional classification networks. The model solution may analyze just the image, which can cut the computational cost of the overall solution by not using an OCR algorithm. Sometimes, however, the image may not give enough information to make a proper decision, so models may also be modeled to classify based on text information, or even mix both in a multimodal solution. This problem is already known and well studied. Bakkali et al. [4] achieved 97.70% accuracy on the RVL-CDIP dataset [5], a popular document classification dataset.

The problem arises when previously accepted documents change in form or when entirely new document classes, not previously considered, must be handled and categorized. In such cases, models are typically retrained, as traditional classification networks may struggle to generalize to new document layouts or entirely new classes. Traditional DL classification methods require a predefined set of labels and cannot output a label outside the original training set. Therefore, introducing new labels to the model means retraining the model, changing the output layer into a wider one with more options. Very often, this also means weeks or months of data engineering, labeling, and model training. The alternative is to use Zero-Shot Learning (ZSL) techniques, which allow models to generalize to classes that were not seen during training [6].

1.2 Problem Description

This work focuses on the ZSL on DIC, where the model must correctly classify a document image class that was not in the training set and therefore previously unseen by the model. ZSL techniques enables models to be considerably more flexible, as they are not limited by the classes used in training. Most ZSL systems work by semantically mapping samples into a feature space, where samples from the same class are all clustered together, and different classes are placed far apart. This is called metric learning, where the model is optimized to map the training samples in regards to a given metric (e.g. euclidian distance). To build a reliable ZSL system, it is necessary to identify a set of features that serve as robust representations, ensuring that documents share features if, and only if, they belong to the same class. With DL, this could also demand a dataset with a wide variety of classes, enough so that the model learns what makes two documents similar or different. These conditions allow ZSL models to group together documents, even if they were completely unseen during training. Nonetheless, not every classification dataset can be easily used in an effective ZSL solution. The lack of a specialized dataset is one of the biggest challenges in document-image ZSL classification.

In consequence, another main challenge in image-based document ZSL classification is the lack of a state-of-the-art methodology. Due to the challenge in achieving ZSL classification with the currently available datasets, most works tackle the problem by their own methodology and very often yield results that are incomparable to their pairs [7], unlike traditional classification and information extraction that have a widely accepted framework and baseline approaches. Many works also use pretrained Large Language Model (LLM) [8], leveraging their previous knowledge as a tool to achieve zero-shot learning with a fine-tuning framework, which limits the cost-efficiency achievable with this strategy.

But there are a few known obstacles in zero-shot document classification. First, existing datasets do not enforce disjoint class splits between training and testing, nor do they provide enough information to train an efficient ZSL model from scratch. Second, there is an absence of a well-established, state-of-the-art classification framework capable of operating under zero-shot constraints.

1.3 Research Objectives

The primary goal of this research is to achieve a complete framework to make a DIC model in a ZSL paradigm, that is, a classification model that does not require retrain every time a new document class is introduced.

To achieve the this, this work pursues the following specific goals:

1. Design and curate a specialized dataset for zero-shot document image classification, ensuring proper class separation between training and testing sets;

2. Develop a framework using metric learning techniques to enable document classification based on visual similarity;

3. Evaluate and compare different NN backbones within the proposed framework to establish performance benchmarks;

1.3.1 Research Hypothesis

This work is grounded on the following hypothesis:

Document images that share similar visual layouts contain discriminative features that can be learned through metric learning, enabling accurate classification of previously unseen document classes by measuring visual similarity, without requiring class-specific training.

This hypothesis is supported by the premise that document layouts within the same class exhibit consistent visual patterns (placement of text blocks, logos, signatures, etc.) that distinguish them from other classes, and that these patterns can be captured through vision models trained with a metric learning framework.

1.4 About this work

Following this line, this work proposes a new framework for tackling ZSL-DIC: Visual Document Matching (VDM). The proposed framework traces de similarity between documents to decide whether or not they share the same class, even with this class has not been seen during training. Through this method, VDM shows itself as a zero-shot alternative for the identification of documents that are visually equivalent, as a matching problem. In other words, VDM can be handled as a binary classification problem.

This work makes several key contributions, i.e.:

1. Introduces a novel document image dataset specifically designed for the task of document ZSL classification, enabling the development and evaluation of models in this domain.

2. Proposes the VDM approach, based on image similarity, leveraging zero-shot learning techniques to generalize across unseen document layouts. This strategy enables generalization to previously unseen classes without additional intervention on the model itself.

3. Delivers a systematic evaluation of well established backbones in the context of zero-shot document layout matching, offering valuable benchmarks for future research.

This dissertation is structured as follows: Chapter 2 lays out the theoretical foundation of this work. Chapter 3 reviews the state of the art in zero-shot learning and visually-rich document understanding, highlighting existing methods and their limitations. Chapter 4 introduces the proposed VDM framework, its architecture and components. Chapter 5 presents the datalabeling process and the experimental setup. Chapter 6 describes the experimental setup, including datasets, evaluation metrics, and baseline comparisons, followed by a comprehensive analysis of the results. Finally, Chapter 7 concludes the dissertation by summarizing the contributions, discussing potential applications, and outlining future research directions, and the timeline of this Master's research.

Chapter 2

Theorical Foundation

The goal of this chapter is to offer the theoretical basis for ideas that are assumed but not elaborated in Chapter 3, Chapter 4 and Chapter 5.

2.1 Traditional Classification

It is mentioned very often in this paper that traditional classification cannot generalize over unseen classes. In the context of this work, it can be specified further: DL models, trained with a cross-entropy learning framework, cannot classify an element into a class that has not been mapped beforehand. This is because the model is trained to minimize the cross-entropy loss function over a labeled dataset, associating each input with a fixed class label. For a single instance, the cross-entropy equation is as follows:

$$CE_{loss} = -\sum_{c=1}^{M} y_c \log(p_c), \tag{2.1}$$

where p_c denotes the predicted probability of the input element belonging to class c, y_c is a label that is 1 if the element belongs to c and 0 otherwise, and M is the number of classes. When a model is trained using a class-disjoint split, meaning that no class appears in both training and test sets, a cross-entropy classifier is unable to learn anything about the unseen classes. Since these classes are not represented in the training data, it is not possible to compute a meaningful cross-entropy loss for them. Therefore, to tackle ZSL, another approach is needed.

2.2 Zero Shot Learning

Unlike traditional classification, which requires the same classes on both the train and the test split, ZSL is the opposite scenario: classes that are in the train set cannot be on the test set [6], i.e., the train and test data split must be class-disjoint. This scenario forces the model to generalize knowledge enough to recognize unseen classes. Generalized Zero-Shot Learning (GZSL) [6] is another similar scenario, but aimed at a more realistic situation. Instead of having a complete class-wise data separation, GZSL proposes that the test split has both unseen classes and known classes. This is intended to simulate a realistic implementation, where a trained model will receive unseen classes as input for inference, but also classes seen during training, as seen in Figure 2.1.

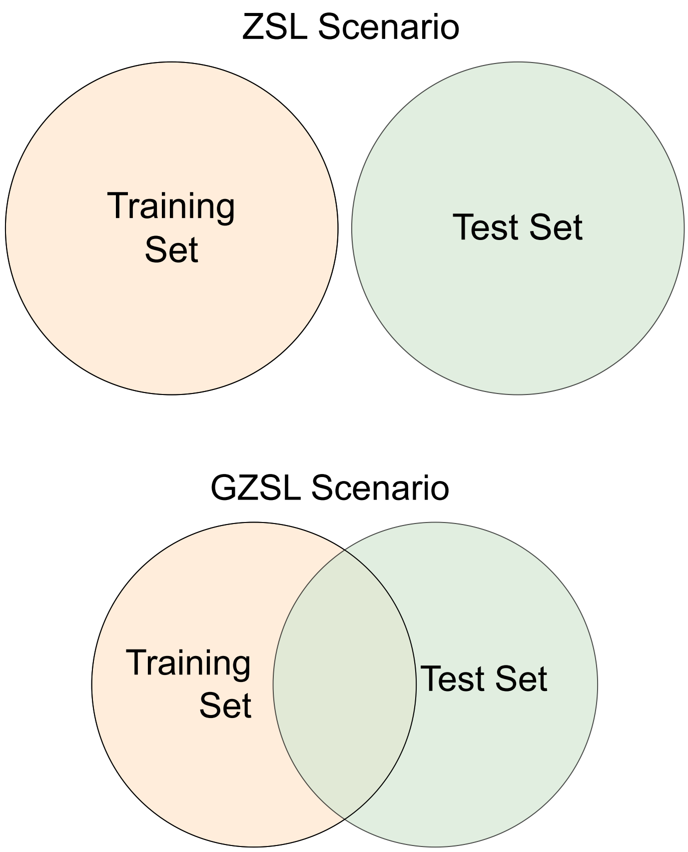

Figure 2.1: Zero-Shot Learning (top) and Generalized Zero-Shot Learning (bottom) scenarios. ZSL has no class intersection on train and test, while GZSL has a partial intersection.

2.3 Metric Learning

One of the possible sulutions to achieve a ZSL classification model, is to use a metric-learning framework. A metric-learning model $f(\cdot)$ is trained such that, over an input element, it yields a feature vector v, instead of a traditional classification [9]. This model $f(\cdot)$ is optimized under a given metric $d(\cdot, \cdot)$, such that two feature vectors that share the same class v_1, v_1' are closer together than two feature vectors from different classes v_1, v_2. In other words, the model is optimized so that $d(v_1, v_1') < d(v_1, v_2)$ and $d(v_1, v_1') < d(v_1', v_2)$.

A DL model following a metric learning framework can be constructed as a siamese network [10]. They are named siamese because they can be seen as bipartite architectures with two parallel paths that converge in the end, such as depicted in Figure 2.2, even though both sides share the same weights. To construct a visual siamese network, an image model, $g(\cdot)$, is required to transform an input image I into a feature representation in an arbitrary n-dimensional space, $g(I)$. The $g(\cdot)$ image model is referred to as the backbone of the siamese network, which is an arbitrary neural architecture (e.g., ResNet [11]).

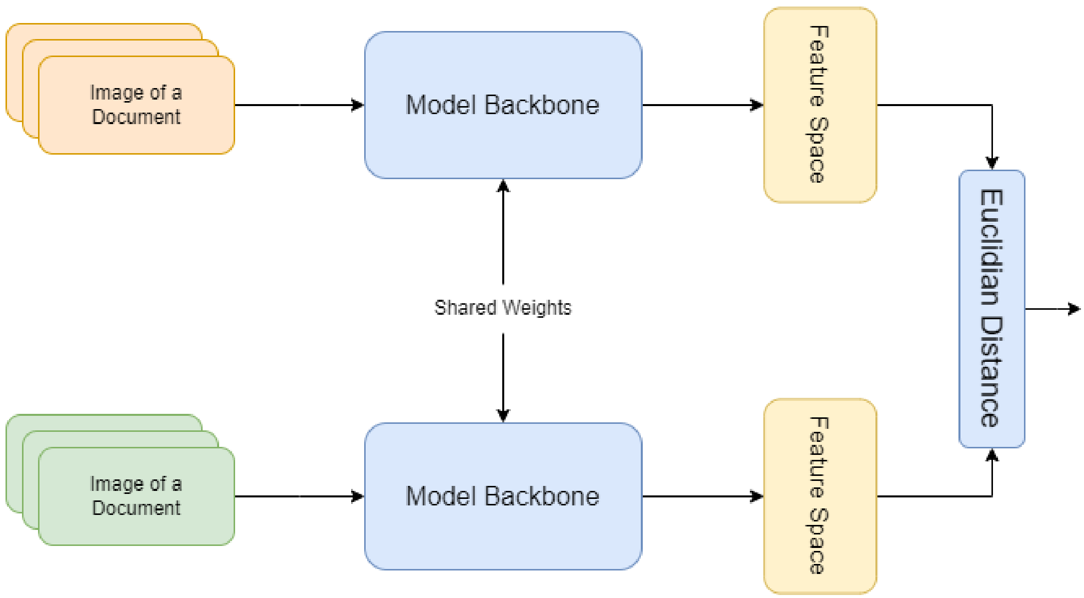

Figure 2.2: Simplified architecture of a siamese network

Once feature vectors are extracted from images, they enable two main inference strategies: verification and identification. Verification addresses the binary problem of determining whether two images belong to the same class. This is accomplished by computing the distance between their feature vectors and comparing them against each other with a predefined threshold, where pairs with distance below the threshold are classified as positive matches (same class), while those above it are classified as negative matches (different classes). Identification, on the other hand, solves the multi-class problem of determining which class a query image belongs to. This is typically achieved through a nearest neighbor search in the feature space, where the query is assigned to the label of its closest neighbor. In this way, verification serves as a binary classifier, while identification functions as a multi-class classifier.

There are many loss functions that support metric learning [12, 13], and they all follow the same strategy: cluster together elements that share a label and split apart elements that do not. When training a siamese network, the training step should consider at least two distinct elements, and the loss function scores the step over the distance between the elements. For example, the contrastive loss [12] considers only two elements per training step, and they can either share the same class or belong to different classes. In this work, the contrastive loss function is used in the experiments.

2.3.1 Contrastive Loss

The contrastive loss function $\mathcal{L}$=$\mathcal{L}(m, y, x_1, x_2)$ is defined as follows:

$$\mathcal{L} = y \cdot d(x_1, x_2)^2 + (1 - y) \cdot \max(0, m - d(x_1, x_2))^2, \tag{2.2}$$

where d is the metric function, $m > 0$ is a hyperparameter defining a distance margin representing the boundary to discriminate what is different and what is similar, and (x_1, x_2) is an input pair. Lastly, y is a label with value 1, if both elements share the same class, or 0 otherwise. Therefore, the loss function can be alternatively expressed as

$$\mathcal{L} = \begin{cases} d(x_1, x_2)^2, & \text{if } y = 1 \\ \max(0, m - d(x_1, x_2))^2, & \text{otherwise.} \end{cases} \tag{2.3}$$

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In this work, we use two different metric functions d: the Euclidean distance and the Cosine distance. The Euclidean distance is defined as the distance between two points $P = (x_{11}, x_{12}, \cdots, x_{1n})$ and $Q = (x_{21}, x_{22}, \cdots, x_{2n})$ in an n-dimensional space, defined mathematically as:

$$d(P, Q) = \sqrt{(x_{11} - x_{21})^2 + \cdots + (x_{1n} - x_{2n})^2} = \sqrt{\sum_{i=1}^{n} (x_1 i - x_2 i)^2}. \tag{2.4}$$

The Cosine distance measures the dissimilarity between two vectors by evaluating the cosine of the angle between them. For two points $P = (x_{11}, x_{12}, \cdots, x_{1n})$ and $Q =$

$(x_{21}, x_{22}, \cdots, x_{2n})$ in an n-dimensional space, the cosine distance is defined as:

$$d(P, Q) = 1 - \frac{\sum_{i=1}^{n} x_{1i} \cdot x_{2i}}{\sqrt{\sum_{i=1}^{n} x_{1i}^2} \cdot \sqrt{\sum_{i=1}^{n} x_{2i}^2}}. \tag{2.5}$$

2.4 Clustering

Clustering techniques are used extensively in this research, as part of the labeling process. Clustering is an unsupervised learning technique that groups data points into distinct clusters based on their similarity, without requiring labeled data [14]. The fundamental objective is to maximize intra-cluster similarity while minimizing inter-cluster similarity, aggregating the data into meaningful groups. In the context of metric learning, clustering becomes particularly powerful as feature vectors learned through metric learning frameworks are specifically optimized to be close together when they belong to the same class and far apart otherwise. This natural alignment between metric learning objectives and clustering goals makes clustering an effective approach for tasks such as class discovery, data exploration, and ZSL scenarios where traditional supervised methods cannot be applied.

K-means [15, 16] is a very popular clustering algorithm. This algorithm begins with k initial seeds, which we call centroids. Then, it follows a two-step iterative process: first, each element is associated with the closest centroid. Next, the centroids are recalculated as the average of the associated elements. This dissertation follows a clustering framework to discover new classes and increase the dataset (see Chapter 5.1), and k-means can be used in this scenario when the number of possible classes in the clustering process is previously known. However, since part of the goal is to discover new, unseen classes, it is not possible to set a k value.

Therefore, a more suitable clustering algorithm is a hierarchical clustering algorithm [17]. Hierarchical clustering builds a hierarchy of clusters through a tree-like structure called a dendrogram, which represents the nested grouping of data points and the similarities at which various clusters are merged or split. There are two main approaches to hierarchical clustering: divisive (top-down) and agglomerative (bottom-up). The agglomerative approach is more commonly used and begins by treating each data point as its own individual cluster. At each iteration, the two closest clusters are merged together based on a chosen linkage criterion.

The chosen linkage criterion significantly impacts the resulting cluster structure. One of the possible strategies is Ward's method [17], which is particularly effective for creating compact, spherical clusters. Ward's linkage minimizes the intra-cluster variance when merging clusters, choosing at each step the pair of clusters on which fusion results in the minimum increase in total intra-cluster variance. Formally, the distance between two clusters C_i and C_j using Ward's method is defined as:

$$d(C_i, C_j) = \frac{|C_i| \cdot |C_j|}{|C_i| + |C_j|} \|\mu_i - \mu_j\|^2, \tag{2.6}$$

where $|C_i|$ and $|C_j|$ are the number of elements in clusters C_i and C_j, respectively, and μ_i and μ_j are their respective centroids. This algorithm supports some stopping conditions, for example, stopping the iterations when a specified number of classes is achieved. A suitable criterion for this work is to stop merging clusters when their distance is greater than m, where m is the margin hyperparameter used in the contrastive loss of the metric learning model, as this model will be optimized treat m as a frontier to what is similar and what is different.

2.5 Vision Models

This section presents the neural architectures used as backbones in this work. All models discussed here are evaluated with the metric learning framework, and their comparative performance analyzed in Chapter 6.

2.5.1 Convolutional Neural Networks

A vision-based DL model is a NN that accepts images as inputs. Vision models can handle a wide variety of tasks, including image classification [18], image segmentation [19], and object detection [20]. Traditionally, these models are based on Convolutional Neural Networks (CNNs) [21], which are neural structures specialized in extracting rich local representations of given data. In vision models, this often reflects on a two-dimensional kernel applied over the input image:

$$(I * K)_{(i,j)} = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} I_{(i+m,j+n)} \cdot K_{(m,n)}, \tag{2.7}$$

where I is the input image, K is the convolutional kernel of size $M \times N$, and (i, j) represents the position in the output feature map.

AlexNet [22], the first NNs to win the ImageNet competition [18], used CNNs as feature extractors to correctly classify images in a traditional classification methodology. AlexNet achieved a 15.3% top-5 error rate in the ImageNet competition, while the second place in that year achieved a 26.2% top-5 error rate, showing the potential of CNN-based models against other classifiers. AlexNet's success rapidly transformed the ImageNet competition, and CNNs became increasingly common. Following this trend, VGG [23] is another CNN-based work that focused its efforts on the ImageNet competition. While AlexNet used only 8 neural layers in total, VGG experimented with increasing the depth of the architecture and its effects on performance. The second biggest model, VGG-16, uses 16 neural layers, and surpassed AlexNet performance with a 8.43% error rate, showing that increasing the depth of a CNN yields better results.

But still, training deep NNs is not as simple as just stacking more layers, as the problem of exploding and vanishing gradient becomes a real impediment [24, 25]. And yet, ResNet [11] is built with even more layers than the previous works. The great breakthrough of this work are the skip-connections, represented in Figure 2.3. Mathematically, instead of learning a desired underlying mapping $H(x)$, the network learns a residual function $F(x) = H(x) - x$, and the final output becomes $H(x) = F(x) + x$. This makes the model easier to optimize, and allows for very deep CNN models to keep learning, even with very deep architectures [11]. The shallowest proposed ResNet with 18 neural layers, ResNet-18, is almost as deep as the deepest VGG, VGG-19. The deepest variant, ResNet-152, achieved a 3.57% top-5 error rate on the ImageNet competition.

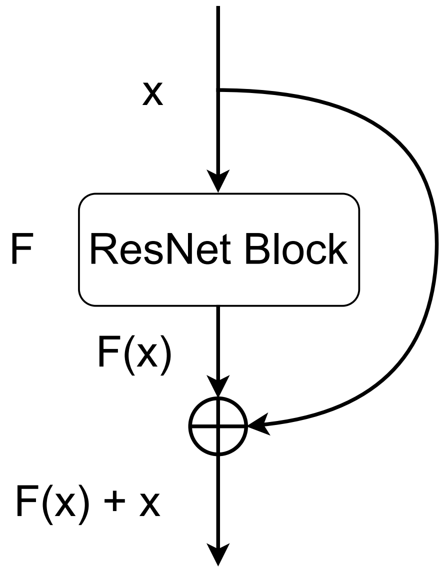


Figure 2.3: Simplified diagram of a ResNet skip-connection.

As CNNs became increasingly more complex, a new wave of works arose aiming to simplify the current landscape. MobileNet [26, 27, 28] is a series of works that progressively achieves better image classification performance while aiming at an economical architecture capable of being executed on mobile devices. MobileNetV3 [28] is the third iteration of this work, and the largest proposed model achieved a 7.8% top-5 error rate in the ImageNet benchmark, with 5.4 million parameters. In the same year, EfficientNet [29] was also published, following the same cost-economic motivation but following a different path. Instead of focusing on devices with low processing power, EfficientNet changes the scaling logic for CNNs. Instead of focusing exclusively on depth, this work proposes to also increase the width (number of filters on each convolution) and the resolution of the input image. This allows the smallest proposed model, EfficientNet-B0, to achieve a 6.7% top-5 error rate with 5.3 million parameters, and the biggest proposed model, EfficientNet-B7, to achieve a 3% top-5 error rate, with 66 million parameters.

2.5.2 Transformers

Transformers [30] is a NN architecture that completely revolutionized the field. The core innovation of transformers is the self-attention mechanism, which allows the model to weigh the importance of different parts of the input when processing each element, enabling the capture of long-range dependencies without the sequential processing constraints of recurrent networks, effectively replacing them. This architecture stands at the top of a wide variety of benchmarks, as the trained models can perform completely new tasks, effectively being able to zero-shot with only natural-language instructions [31].

FIGURA VIT

This revolution, for a while, stayed concentrated in Natural Language Processing (NLP) research, while vision research remained with ResNet-like architectures as state of the art [32]. Then, there was ViT [32]. This work takes the Transformers from NLP research and makes an effort to use it in an image context, making the least amount of adaptation as possible. To achieve this, Vision Transformer (ViT) applies this architecture by dividing an input image into fixed-size patches, linearly embedding each patch, and processing the sequence of embedded patches through transformer encoder layers. This approach treats image patches analogously to tokens in NLP, allowing the model to learn global relationships between different regions of an image. Despite requiring large amounts of training data to achieve competitive performance, ViT-based models have achieved state-of-the-art results on various vision benchmarks, usually by pre-training the model on a large dataset and then fine-tuning to solve a specific problem.

Table 2.1 provides an aggregation of the discussed vision models and their performance on the ImageNet classification task [18]. The performance metrics are taken from their own work. AlexNet [22] and VGG [23] do not report top-1 error rate, while ViT does not report top-5 error rate. Note that the models presented in this table are all used in this research, along with a few other variants.

Table 2.1: Comparison of popular neural network architectures in computer vision.

Model	**Year**	**Parameters (M)**	**ImageNet Performance**
AlexNet	2012	60	Top-5 err. 15.3%
VGG-16	2014	138	Top-5 err. 7.3%
ResNet-152	2015	25.6	Top-1 err. 19.38%, Top-5 err. 4.59%
MobileNetV3-Large	2019	5.4	Top-1 err. 24.8%, Top-5 err. 7.8%
EfficientNet-B0	2019	5.3	Top-1 err. 22.9%, Top-5 err. 6.7%
EfficientNet-B7	2019	66	Top-1 err. 15.7%, Top-5 err. 3.0%
ViT-B/16	2020	86	Top-1 err. 11.63%

Chapter 3

Related Work

This chapter analyzes the literature about automatic document processing, focusing on the available datasets and document classification methods. The relevance and direction of this work are justified by the limitations of each dataset and classification framework. This chapter also helps to provide the boundaries for the problem.

3.1 Document Understanding

Document Understanding is a broad field that acts as an umbrella to a wide variety of document-related tasks, and its goal is to transfer physical information, presented in paper, to a digital version that can be stored in databases and used in computer processes [33]. The increasing digitization of documents has led to significant advancements in document analysis techniques, particularly through DL and transformer-based architectures [34]. Among the key challenges in this domain is handling the inherent complexity of scanned documents, which often exhibit noise, structural variability, and diverse formatting styles.

Several fundamental tasks define the scope of document understanding. Document Layout Analysis involves detecting and categorizing different structural components of a document, such as text blocks, tables, images, and forms, to facilitate higher-level information extraction [35]. Information extraction focuses on retrieving relevant information, such as named entities and key-value pairs, from structured and unstructured document sources [34]. This work focuses on DIC, which focuses on correctly classifying a document image into a class. It is important to point out that document image tasks do not provide the document class as input for the experiments, and if the text is required to the proposed solution, it needs to be extracted from the image, for example, using OCR solutions.

3.2 Document Understanding Databases

This chapter reviews existing datasets commonly used for document classification and discusses their limitations in supporting zero-shot scenarios. By analyzing these datasets, the need for a new benchmark that explicitly enforces zero-shot constraints is highlighted. Since the document understanding field is so broad, there are many document-image datasets available. This chapter goes through many of them and points out the limitations they show when seen through the ZSL-DIC optic.

PubLayNet [35] and DocLayNet [36] are large-scale datasets specifically designed for document layout analysis tasks. PubLayNet contains over 360,000 document images, with labels for five layout categories: text, title, list, table, and figure. DocLayNet extends this scope by providing 80,863 manually annotated pages from more diverse document sources, categorized into 11 layout classes, instead of 5. While these datasets have significantly advanced document layout analysis research and serve as foundational resources for training models to detect and segment structural components, they are not designed for document classification tasks. Their labels include pixel per pixel class annotations, making them unsuitable for directly training or evaluating DIC approaches.

More recently, the DocVQA dataset [37] was introduced, leveraging a subset of documents from the same collection. It comprises over 12,000 document images paired with 50,000 question-answer pairs, designed to evaluate models' abilities in visual question answering on document images. DocVQA has since become a standard benchmark for assessing the performance of LLMs in document understanding tasks, particularly in multimodal reasoning scenarios. FUNSD dataset [38], introduced in 2019, has been widely used for OCR-based semantic relation extraction from scanned forms. Similarly, SROIE [39], CORD [40] and XFUND [41] target key-value pair extraction in receipts and invoices, focusing on structured entity retrieval, such as store names, monetary values, and transaction dates. All these three datasets focus on, in one way or another, extracting specific information from documents. Again, these labels cannot be directly used to train a classification model, as they do not provide a class-based separation, nor a class-disjoint split we expect in a ZSL dataset. Figure 3.1 shows examples of document layout analysis and information extraction datasets.

At the end of the day, the most suitable datasets for ZSL-DIC task are the traditional classification datasets. The biggest source of documents for these datasets is the IIT-CDIP dataset [42], introduced in 2006, which is a large-scale repository used for document classification and information retrieval. It originated from the Tobacco Documents Library, from the University of San Francisco Industry Documents Library collection, and contains millions of scanned documents. The most popular classification dataset derived from the IIT-CDIP is the RVL-CDIP dataset [5], introduced in 2015. This dataset is a

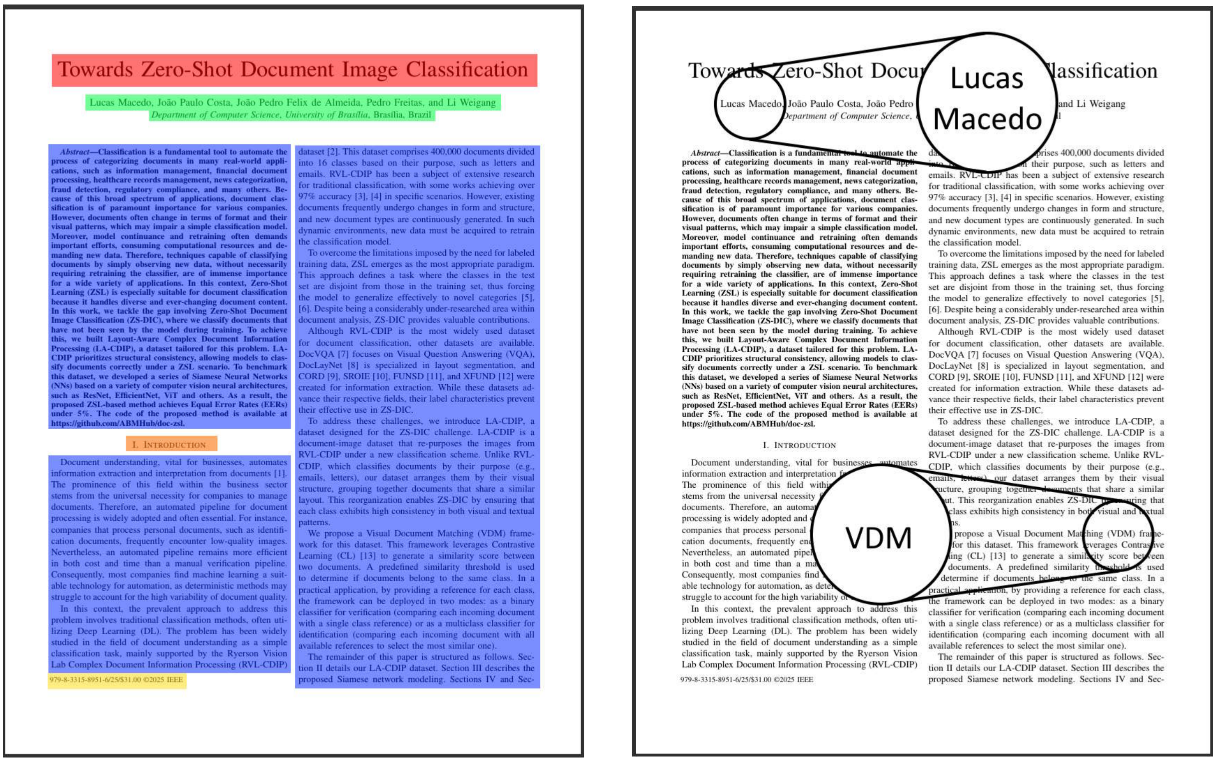


Figure 3.1: Examples of document understanding datasets. Left illustrates a document layout analysis dataset, where the goal is to visually segment the layout information. Right illustrates a information extraction dataset, where the goal is to find specific informations about the document, i.e., the author name.

labeled subset of the IIT-CDIP dataset and organizes 400,000 document images into 16 predefined categories such as letters, forms, and emails. This classification is driven by the document's purpose and uses, which hinders the performance of a ZSL model trained from scratch, with no external real-world knowledge. This observation is confirmed by, Larson et al. [43], who highlighted this difficulty by stating that each Ryerson Vision Lab Complex Document Information Processing (RVL-CDIP) class comprises various subclasses, as illustrated in Figure 3.2. They further argue that this may constrain ZSL solutions, as no single reference could specify an entire class without overlaps. Other traditional classification datasets suffer from the same problem, as they share very similar classes and labeling logic [44].

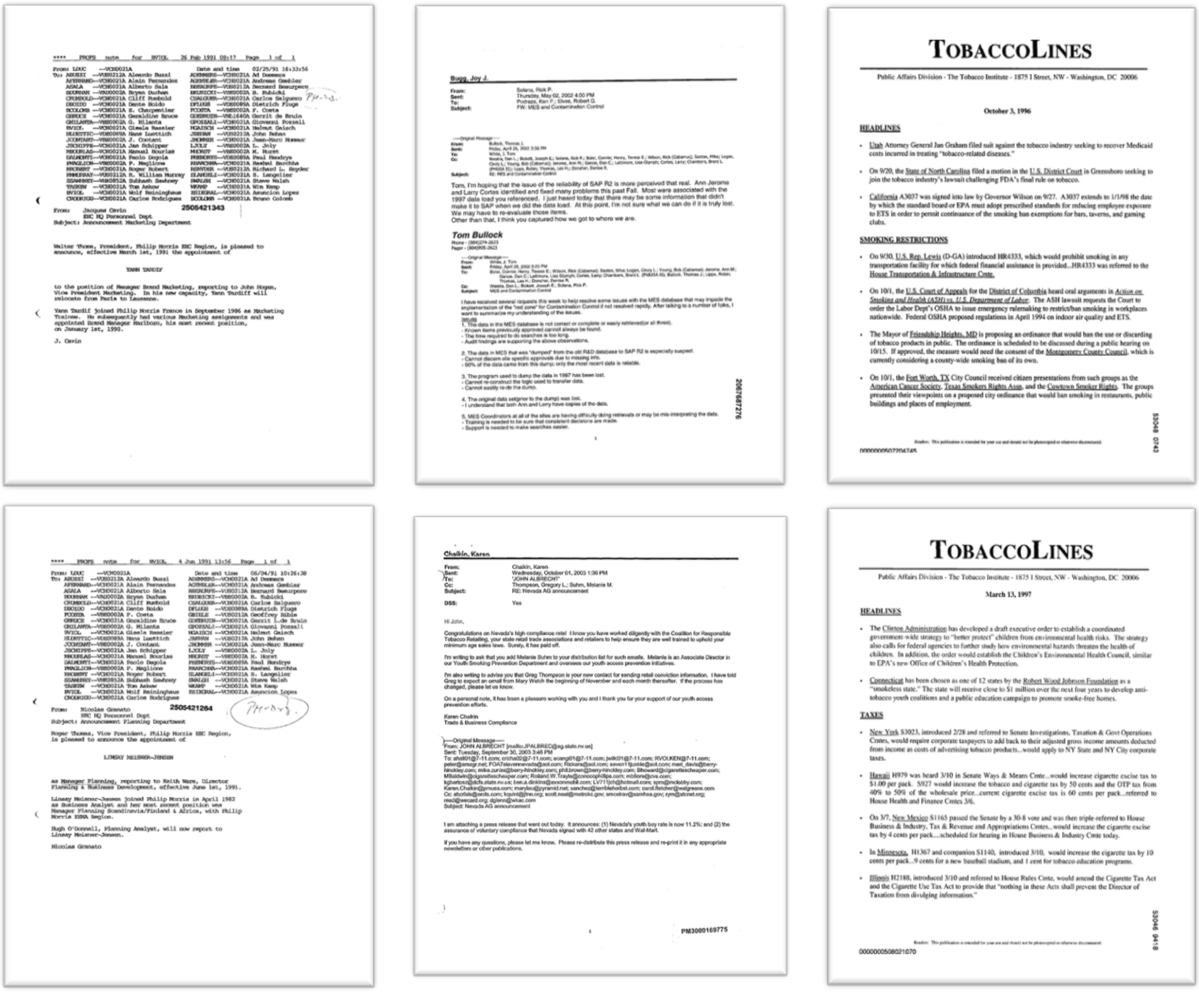


Figure 3.2: Six examples of the RVL-CDIP dataset. Note that, despite the visual differences, these six documents all share the same class under the RVL-CDIP organization.

Therefore, even with these available datasets, there is still a gap that can be explored in the ZSL-DIC domain. The existing datasets either lack the appropriate structure for classification tasks or present challenges in zero-shot scenarios due to overlapping class definitions and the absence of explicit class-disjoint splits. This limitation motivates the need for a new benchmark that explicitly addresses these constraints and enables more robust evaluation of zero-shot document classification approaches.

3.3 Zero-Shot Document Classification

The ZSL-DIC field is historically underdeveloped. One of the possible main reasons is the lack of a proper dataset. Even then, the research community took some different approaches to ZSL-DIC. Most of the research is done with RVL-CDIP dataset, even with the presented limitations.

Khalifa et al. [45], using the RVL-CDIP dataset, proposed a matching solution for ZSL-DIC. Using contrastive learning, they encode both the document itself and the label string in a metric learning approach. They achieve a 39.63% F1-score in an unsupervised setting, that is, with no task-specific training, using exclusively pre-trained models. Then, they train the models in a contrastive-learning framework and achieve 37.23% F1-score. Note that the performance actually decreased. They mention that training the models in a subset of the classes in RVL-CDIP, then testing with the remaining classes, hinders the generalization on the unseen classes. They close the paper suggesting that the RVL-CDIP dataset is not suited for the ZSL-DIC problem, and that a new dataset must be created to continue the research, arguing that the RVL-CDIP classes are comprised of various subclasses.

Scius et al. [8] investigate the application of LLMs, such as GPT-4 [46] and RoBERTa [47], for zero-shot prompting and few-shot fine-tuning in document image classification. Focusing on their zero-shot experiments, their framework is based on describing to the LLM, through a prompt, about the classification task and how it should classify each document. Their study demonstrates that LLMs can achieve competitive performance with minimal labeled data, challenging the traditional reliance on large annotated datasets. Their best performance with open source LLM achieved an average of 54.4% accuracy, using a model of 123 billion parameters, and achieved 69.9% accuracy with GPT-4-Vision, which has an undisclosed parameter count.

Based on Khalifa et al. [45] work, Sinha et al. [7] introduce CICA, a framework that enhances CLIP's [48] performance in zero-shot classification by joining it with a content-encoder, which aligns its own feature representations with the feature space used by CLIP in a contrastive-learning approach. CICA manages to bypass the RVL-CDIP limitations by exploring CLIP's pre-trained knowledge, achieving a 67.29% average accuracy. Compared to the previous works, this study emphasize data-efficient approaches to document classification. However, their approach is not suitable for a from-scratch training, which would allow an even greater efficiency gain.

Chapter 4

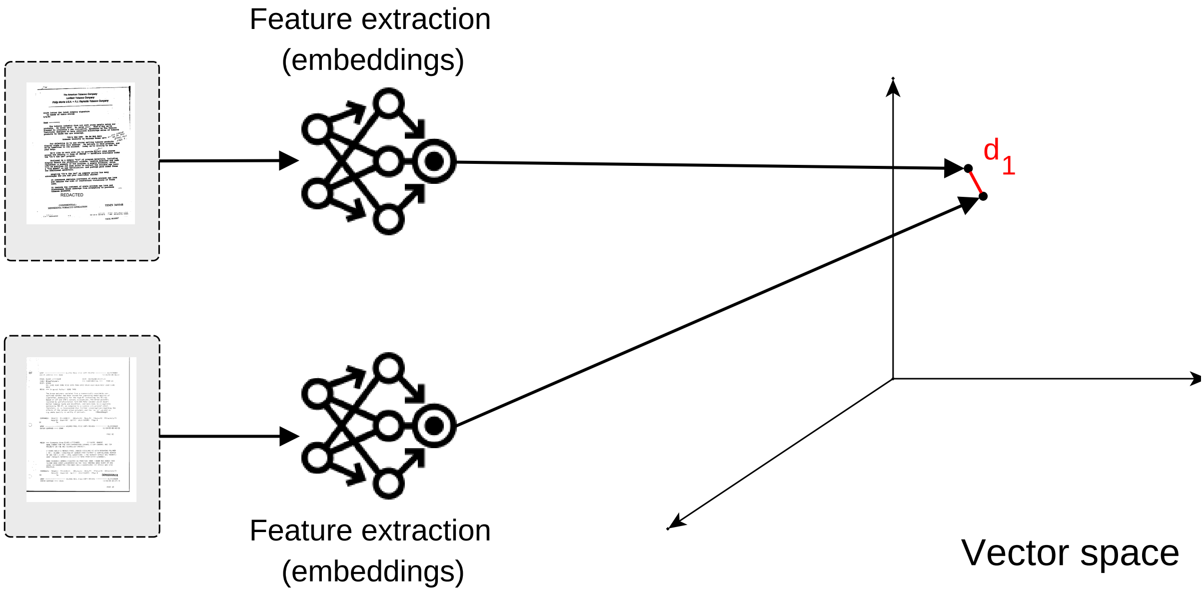
Proposed Solution

4.1 Visual Document Matching

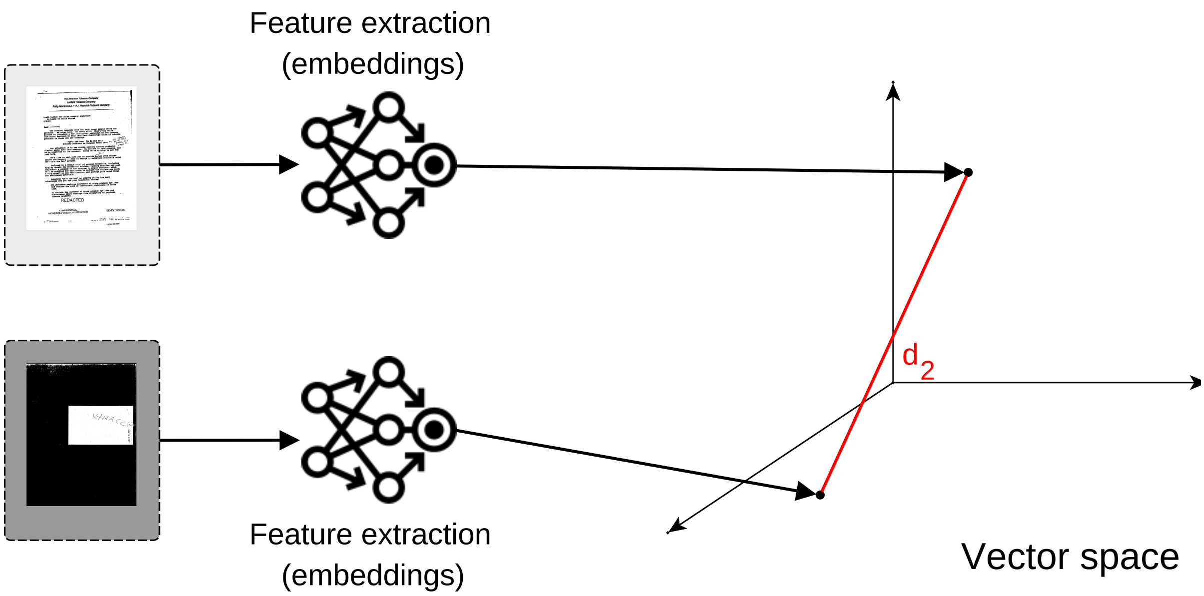
Traditional classification methods, such as entropy learning, lock the generality of the model to the set of classes it has been trained on. Therefore, the use of Zero Shot Learning techniques are essential to be able to classify document layouts that have not been seen on the training set. Among the possible ZSL solutions, we use metric learning with siamese networks to achieve this goal. The proposed metric learning framework, VDM, takes two different document images, and compare them to decide whether they belong to the same class or not, using vision models to match documents that share the same class.

This approach is inspired by biometric signature verification approaches [49], where a system has a repository of ground-truth signature references (signatures that they know are from certain person), and they compare them with a query signature, to verify if they belong to the same person or not. Therefore, VDM works by learning to draw representations of documents in a feature space with n dimensions, such that documents that share the same class are represented clustered together, and documents that have different classes are far apart, as illustrated in Figure 4.1.

A wide variety of backbones are used to experiment on the proposed dataset, but they all follow the same metric learning architecture with Siamese Networks. They are AlexNet [22], VGG [23], ResNet [11], MobileNetV3 [28], EfficientNet [29] and ViT [32]. To adapt the architectures to a siamese network, only the last linear layer of each model—the classification layer—is modified into a new linear layer with an arbitrary size n, suitable for the problem.



(a) Similar documents with small distance d_1 between them.



(b) Dissimilar documents with large distance d_2 between them.

Figure 4.1: Illustration of two documents mapped to the vector space that are similar (a) and dissimilar (b). This document mapping to a vector space is performed by the proposed learned method for similarity analysis. Visually similar documents have a smaller distance in the projected space than dissimilar ones.

Chapter 5

Methodology

5.1 LA-CDIP Dataset Construction

The construction of this dataset is motivated by the absence of a specialized ZSL-DIC dataset, and the fact that RVL-CDIP is unsuitable for the proposed VDM method [45] (see Chapter 3.3). To support VDM in achieving ZSL-DIC, this dataset must provide two main characteristics: the dataset must provide great class diversity, so that models trained on this dataset can generalize across unseen classes. And the dataset must have consistent intra-class and inter-class consistency, so that the classes are atomic (with no possible subdivision).

Therefore, the proposed dataset, Layout-Aware Complex Document Information Processing (LA-CDIP), comes as reorganization of the RVL-CDIP dataset, focusing on increased granularization, in the sense that no two document layouts are seen on the same class. Visual cues such as a company logo, the layout of a table, the position of certain text, etc., all contribute to differentiating between distinct document patterns, and different patterns imply different classes. While the RVL-CDIP dataset was used as source content, it was re-labeled to align with the problem of differentiating documents, with a particular focus on their visual structure (layout). Figure 5.1 illustrates the impacts of the new labels. All eight documents in the figure used to belong to the same class, but now belong to two different classes under the new organization.

The dataset labeling following two stages: preliminary clustering, followed by a manual reorganization. The clustering is innitially made with a private metric learning model, trained with a private document dataset, with both the model and the dataset following the same methodology as this dissertation. Following, the model is used to generate feature vectors across the entirety of the RVL-CDIP. These feature vectors are then used to create clusters of documents, which are effectively predicted classes. The Hierarchical

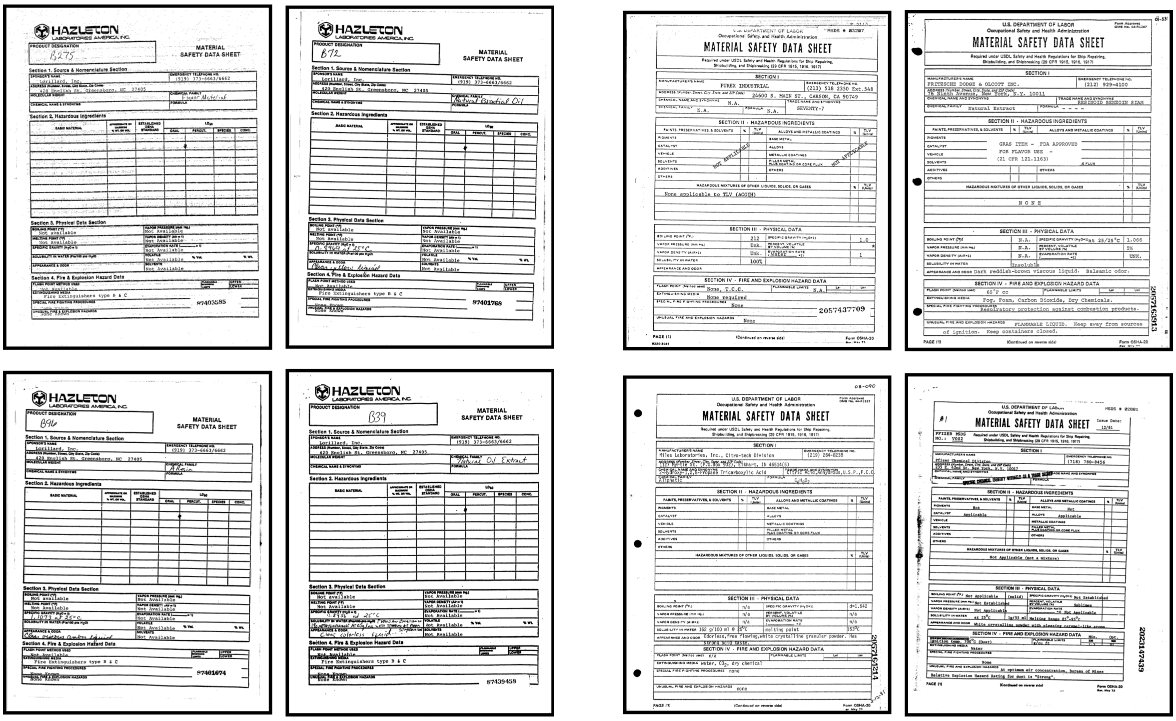


Figure 5.1: Examples of two distinct classes under the proposed LA-CDIP dataset. Those classes originally belonged to the same class under RVL-CDIP organization.

Agglomerative Clustering algorithm, with Ward's[17] strategy is used (see Chapter 2.4). This clustering served the purpose of accelerating the manual labeling process.

The manual labeling process has two main tasks: first, verify the integrity of the clusters. As the clusters were made with a DL model, they are prone to errors. Therefore, it is possible that one cluster holds many different document patters, which must be corrected by a human annotator, assuring intra-class consistency. Second, verify if this cluster is unique. The model may have created two distinct clusters with the same layout, therefore, a step to merge clusters is necessary, assuring inter-class consistency.

This training pipeline is bound to a loop: given stop condition (which can be a time frame, or a goal in number of labeled documents), the annotated documents are comprised into a dataset, with more elements than the previous iteration, and are used to train a new model, using the same VDM framework. This new, enhanced model is then used to create new clusters, which should have a greater intra-cluster and inter-cluster consistency, therefore, reducing the human annotation time necessary per document. To further optimize the labeling process, each cluster is ranked based on average pairwise distance between every element in the cluster, and the clusters with lowest average distances are provided first to the human annotators. This is meant to optimize the amount of work to achieve intra-class consistency.

Note that the proposed data labeling loop is similar to active learning [50], with a

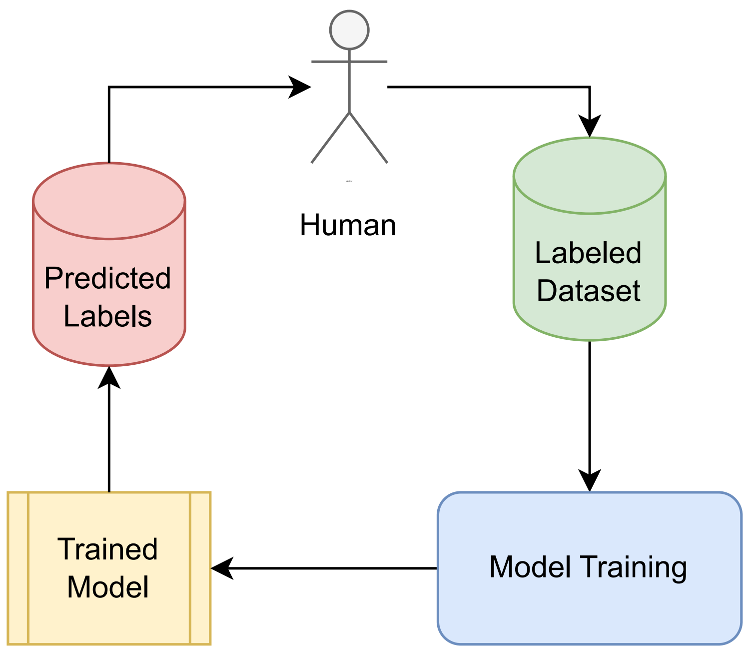


Figure 5.2: The proposed data labeling loop. Each loop, the labeled dataset is increased, which is used to train a better model, which results in more consistent predicted labels, reducing the time the human annotator spends per document.

few key differences. While the LA-CDIP dataset is created to support VDM, it is still a contribution by itself. Therefore, the purpose of this labeling loop is to create a large and complete dataset, which differs from active learning philosophy, which is to achieve greater accuracy with fewer labeled instances [50]. The sampling strategy frequently used in active learning involves focusing the human efforts in samples that the model did not provide confidence in its classification, so that they provide the most value. This is opposite to what is proposed in this work, which is to give priority to clusters that have the most confidence, optimizing the time spent per annotation.

5.2 LA-CDIP Dataset

The proposed dataset is composed of 4993 documents, divided into 144 different classes. Each class has at least two documents, the biggest class has 497 documents, and the median size of the classes is 13, exposing an extra challenge in dealing with an unbalanced dataset. Figure 5.3 shows the class distribution.

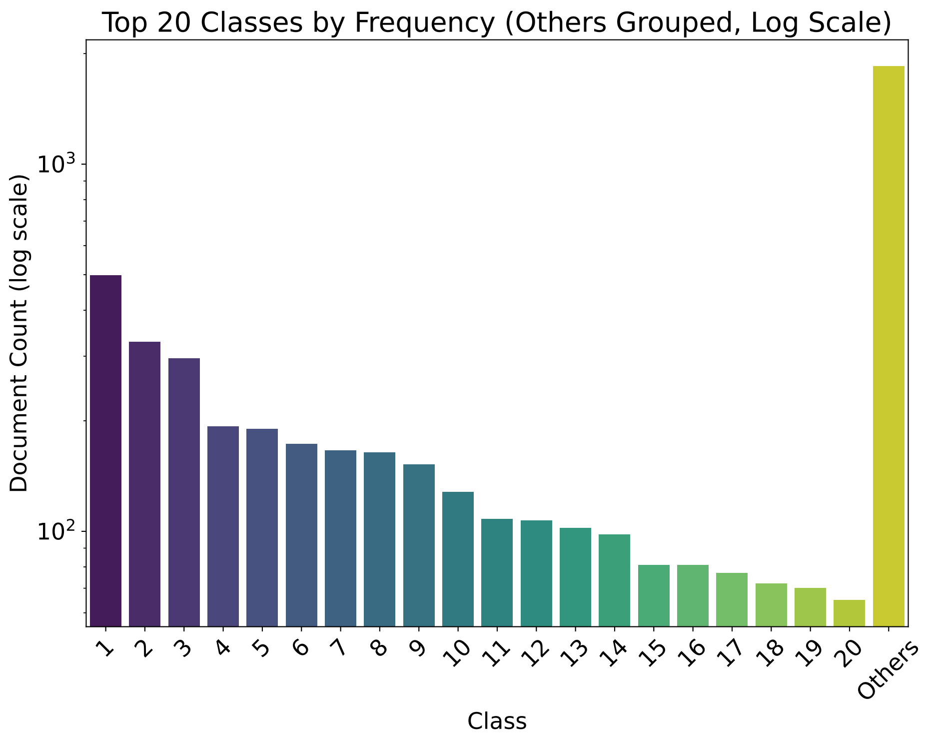


Figure 5.3: Histogram showing the top 20 classes by their frequency. The other 124 classes are grouped in the "others" column. Note that the plot is in log scale.

To maintain train-test data consistency, the data is split following the ZSL and GZSL protocols proposed by Xian et al. [6] (see Chapter 2.2). In both scenarios, the data is split into train and test data, and the train data is further divided by a 5-fold cross-validation [51]. The data for the test scenario and the data for each split are chosen randomly, and remain the same for every experiment. The test scenario is picked as a sixth split for the cross-validation, so it follows the same rules as the other five splits. For the ZSL scenario, one-sixth of the classes are randomly chosen for each split, with no overlapping. Naturally, this scenario creates splits with variable sizes, as the classes themselves have variable sizes. For the GZSL scenario, half the classes in the whole dataset are chosen as classes that do not overlap between splits, and the other half are distributed across the splits. While constructing a split, first the nonoverlapping classes are chosen for the split, then the remaining slots are filled with the overlapping classes, achieving splits with a constant size for this scenario.

This dataset can be normally used for a traditional visual classification problem, but since the problem is tackled with metric learning, additional considerations must be taken into account. While using a Siamese network, running inference on a single point of data returns no information, as the architecture requires some sort of comparison between different data points. Randomly selecting pairs every test can fluctuate the results, therefore, to maintain test consistency, a test protocol is used that ensures the same pairs are tested every time. For every document in the set, two other documents are chosen: one that shares the same class, and one that does not. This document is chosen randomly by its class, therefore, every class has the same probability of being picked. A protocol exists for the test set and for every cross-validation split.

5.3 Evaluation Criteria

Since a metric learning model yields a distance between two data points (in this case, two documents), a threshold needs to be defined to declare whether the two data points belong to the same class or not, to calculate how accurate the model is. For this, Equal Error Rate (EER) is used as the metric for the experiments [52]. EER is the point at which the False Acceptance Rate (FAR) and False Rejection Rate (FRR) are equal. The EER is calculated by setting the FAR equal to the FRR and finding the corresponding threshold at this point [53]. In other words, the EER is the value of error rate when the threshold value τ_{EER} gives

$$FAR(\tau_{EER}) = FRR(\tau_{EER}), \tag{5.1}$$

where

$$FAR(\tau) = \frac{\text{Number of false acceptances at threshold } \tau}{\text{Total number of negative samples}} \tag{5.2}$$

is the probability that a negative sample is incorrectly classified as positive. On the other hand,

$$FRR(\tau) = \frac{\text{Number of false rejections at threshold } \tau}{\text{Total number of positive samples}} \tag{5.3}$$

is the probability that a positive sample (e.g., a genuine user) is incorrectly classified as negative.

Ten models are trained for each chosen backbone (see Chapter 4.1): a 5-fold cross-validation for each of the 2 scenarios, ZSL and GZSL. As shown in Section 5.1, each split follows a fixed validation protocol to ensure consistency. The trained models from each split are also evaluated on the independent test set, to compare them with the LLMs. The mean EER is reported for every cross-validation scenario, as well as the mean EER on the independent test set.

5.4 VDM Experimental Setup

The method was implemented in Python language, mainly using the PyTorch library [54]. The hyperparameters used in these experiments are shown on Table 5.1. Note that, while most of the architectures use the same hyperparameter, MobileNetV3 and ViT use different configurations, following suggestions from their original works.

Table 5.1: Hyperparameters for different neural architectures in VDM experiments.

Architecture	**Optimizer**	**LR**	**LR Decay**	**WD**	**Epochs**
Default	SGD	0.01	Step (0.1/30 ep.)	1×10^{-4}	90
MobileNetV3	SGD	0.01	Step (0.973/2 ep.)	1×10^{-5}	300
ViT	AdamW	0.001	Cosine Annealing	1×10^{-5}	90

The VDM trained models are exclusively visual; therefore, their input data is the (R, G, B) document image matrix. First, the input image is resized into a shape compatible with each neural architecture. For most models, this shape is $(224, 224)$ in height and width. EfficientNet is the only architecture that does not follow this rule, as the different model versions increase their input size at the same time they increase the network depth and width. Then, every value of the image matrix is scaled from 0–255 to 0–1, and normalized with the mean and standard deviation of the current training split.

During training, each step processes a pair of documents that can be either similar (same class) or dissimilar (different classes). The pairing process follows a specific procedure: first, a random document is selected from the training set; then, the pair type (similar or dissimilar) is randomly determined. For similar pairs, another document from the same class is chosen. For dissimilar pairs, a different class is randomly selected first, followed by a random document from that class. This class-level randomization ensures that every class has equal probability of forming dissimilar pairs, preventing overfitting to larger classes. To maintain balance during training, half of all pairs are similar and half are dissimilar. Each batch is composed of 16 pairs. Throughout an epoch, each document is chosen twice to compose the first position of the pair: once to make similar pairs, and another to make dissimilar pairs. While the loop training randomizes each pair each epoch, during test and validation, the pairs are fixed. The chosen pair is then processed by the siamese network, generating two feature vectors, which are evaluated by the contrastive loss function (see Chapter 2.3.1).

Chapter 6

Results

6.1 Partial Results

		Cosine					Euclidean				
Model	Version	16	32	64	128	256	16	32	64	128	256
AlexNet	Base	8.11	8.30	5.79	6.37	7.92	15.64	13.90	20.66	18.73	21.62
VGG	11	6.37	10.04	9.46	14.09	12.55	8.88	15.06	19.88	17.76	15.06
	13	10.81	11.00	6.76	7.53	8.69	15.25	17.37	12.55	17.18	18.15
	16	9.65	7.92	7.53	8.49	11.78	13.71	8.49	10.81	8.69	10.42
	19	7.92	12.55	7.53	5.98	16.41	17.95	16.80	9.85	16.41	20.27
ResNet	18	7.34	6.76	5.60	4.63	6.95	3.86	5.79	4.83	7.34	5.98
	34	7.53	4.63	5.79	3.28	5.98	7.53	11.40	7.53	12.16	9.85
	50	11.58	7.14	6.95	9.07	11.20	6.95	7.14	9.85	8.69	17.57
	101	6.37	11.58	5.79	5.98	5.79	6.18	5.98	10.04	13.32	6.95
	152	7.14	5.79	7.53	5.21	5.79	5.21	9.46	17.37	7.92	19.31
MobileNet	Small	8.69	8.49	16.99	14.67	9.07	9.07	17.18	7.92	10.81	12.93
	Large	10.81	10.04	8.11	10.81	8.49	11.20	14.86	9.27	7.14	13.32
EfficientNet	0	9.27	11.20	11.00	9.85	10.81	7.92	6.95	5.02	4.05	4.83
	1	7.92	6.95	7.14	5.21	5.60	5.02	8.69	8.69	10.04	11.58
	2	12.93	12.55	10.81	11.58	10.23	7.92	8.49	8.30	7.34	5.41
	3	8.11	9.65	9.27	5.60	8.49	5.41	5.60	6.37	8.49	11.39
ViT	Base	20.65	21.05	21.24	20.65	18.73	19.88	17.37	20.27	19.69	19.69
	Large	24.13	24.52	22.97	31.47	20.27	20.08	21.43	17.37	25.48	17.76

The results of the research are presented in this chapter. This chapter's discussion revolves around Table 6.1, where the results of every trained model are presented. The table shows the mean EER value of the cross-validation in both scenarios, as well as the mean performance of each fold on the independent test set. While the ZSL and GZSL

		Cosine					Euclidean				
Model	Version	16	32	64	128	256	16	32	64	128	256
AlexNet	Base	2.10	2.10	1.85	2.10	1.98	4.70	9.77	8.65	11.37	12.36
VGG	11	3.46	2.60	3.46	3.09	4.94	4.08	5.32	6.30	6.43	7.05
	13	2.72	2.47	2.60	2.47	3.58	8.15	5.32	5.32	8.28	6.30
	16	1.73	0.99	2.97	1.61	2.72	6.30	3.96	4.45	6.18	4.58
	19	3.09	2.22	2.47	0.87	2.60	3.58	7.17	5.44	3.96	8.65
ResNet	18	1.36	0.99	0.74	0.37	1.11	3.71	2.35	3.46	5.44	2.84
	34	1.73	0.87	1.85	0.62	0.74	2.84	3.71	4.33	3.09	3.58
	50	2.72	1.85	0.99	1.24	1.61	6.67	3.21	4.70	4.33	6.67
	101	2.97	2.47	2.10	1.98	1.98	3.21	4.58	2.97	7.54	8.65
	152	2.72	2.72	1.85	2.35	2.47	6.67	3.96	5.69	5.81	5.81
MobileNet	Small	6.43	2.84	4.33	5.69	4.08	6.43	8.41	5.69	5.93	9.27
	Large	4.45	4.82	4.58	5.44	5.07	7.54	5.69	6.18	5.81	7.42
EfficientNet	0	1.24	1.61	0.49	0.87	1.73	2.10	2.10	2.97	3.96	3.96
	1	1.61	1.24	0.62	1.24	1.98	2.22	3.09	3.33	2.47	2.84
	2	1.73	1.24	1.24	1.61	0.99	1.36	2.47	2.84	4.33	3.58
	3	1.85	1.48	1.36	1.24	1.36	1.24	0.99	3.21	4.82	5.19
ViT	Base	14.96	14.83	10.88	14.22	14.71	10.26	8.78	10.51	9.64	9.64
	Large	16.07	17.68	14.83	22.37	12.73	10.88	11.37	8.78	14.09	10.51

scenarios are very influential on the visual models' performance, this difference is not relevant in the LLM test, as they have not been fine-tuned for the task. While they may have been originally trained with some documents of the RVL-CDIP database, both scenarios are effectively ZSL. Chapter 6.1.1 discusses the results of the trained visual models and compares the effects of different backbones on the result. Chapter 6.1.2 compares the results returned by the multimodal LLMs chosen for this work. Finally, Chapter 6.1.3 compares both approaches.

6.1.1 Visual Models

Among visual models, smaller and more cost-efficient architectures outperformed larger ones. This trend is evident in different ResNet variants: ResNet-18 and ResNet-34 showed strong performance, while larger versions (ResNet-50, ResNet-101, and ResNet-152) performed progressively worse. Additionally, the ViT, despite excelling on datasets like ImageNet [18], was the worst-performing model in this task, both architectures (ViT-b and ViT-l) achieving over three times the amount of error, in comparison with EfficientNet-b1 in the ZSL scenario. These larger models overfitted to the training data, with some reaching 0% train error in certain epochs while maintaining a high validation error. The most likely cause of this behavior is the dataset size—LA-CDIP contains only 4,993 documents, with approximately two-thirds used for training in each fold. However, techniques such as tuple mining and data augmentation could help mitigate this effect by increasing the diversity of training samples and improving feature learning. The overall best performing models on this benchmark are the ResNet-18, achieving a 1.54% error rate on the GZSL scenario, and EfficientNet-b0, achieving a 3.93% error rate on the ZSL scenario. Both models offer a balance in size, achieving a good generalization of the problem while also learning the intricacies the task demands.

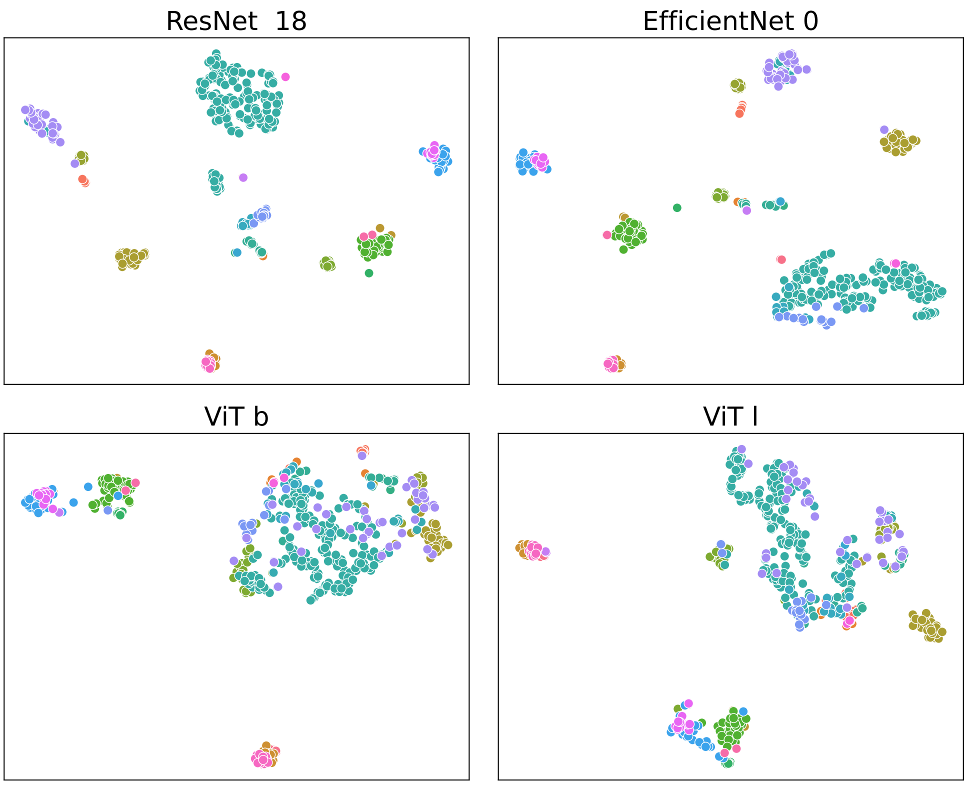


Figure 6.1: TSNE visualization of the Test ZSL scenario. These models were trained on the same cross-validation fold. Each dot represents a document positioned in this two-dimensional space through TSNE dimensionality reduction. Each color represents a different class.

Comparing the ZSL and the GZSL results, the GZSL consistently gives better results. This is expected, since half of the validation set on each fold is composed of classes seen in training. The biggest discrepancies between the two modalities can be seen in the largest models, where the overfitting damage is mitigated by the easier scenario. Even so, the best performance on the GZSL scenario is usually connected with a good performance on the ZSL scenario.

In the test scenarios, the reported values represent the mean performance across all folds, evaluated on the same set. Since this is a Zero-shot data split, there is inherently a high variance over the different splits, since they test on entirely different patterns each fold. By chance, the ZSL test split is more challenging than average, resulting in higher error rates for most models relative to the mean cross-validation error rate. The test GZSL scenario follows the opposite reasoning.

6.1.2 Large Language Models

All models were tested under similar conditions; however, further performance improvements could be achieved through refined prompt engineering. Therefore, the results presented should not be considered definitive comparisons of the models' capabilities but rather a baseline for assessing the viability of the proposed method. Despite these limitations, LLM results align with existing benchmarks for document understanding tasks for these models.

As shown in Table 6.1, GPT-4o exhibited the best overall performance, achieving the lowest error rates in both the ZSL and GZSL scenarios. The GPT-4o-mini variant performed similarly but showed slightly lower accuracy, reflecting its trade-off between efficiency and capability.

Additionally, Llama 3.2 Vision faced challenges in handling multiple document inputs, requiring images to be combined into a single input before processing. This limitation impacted its ability to directly compare layouts across distinct documents, differentiating its approach from the other evaluated models.

InternVL 2.5 and Qwen2.5-VL emerged as open-source alternatives for ZSL VDM. Notably, the QwenVL 7B model reached performance levels close to those of GPT-4o mini, further highlighting the efficiency of these compact architectures.

Since the LLMs have not been fine-tuned with the training data, the only difference between the ZSL and GZSL scenarios is the number of classes to compare: ZSL has exactly $\frac{1}{6}$ of the classes and GZSL can include every class, making GZSL a more diverse test. Surprisingly, the two models performed with different trends between both scenarios: InternVL was better in the ZSL split, and GPT better in the GZSL split.

6.1.3 Comparison

Given that GZSL is an advantageous scenario for vision models—since they have been trained on half of the classes—while LLMs have not been fine-tuned for this task, both settings effectively serve as pure zero-shot evaluations. Therefore, the comparison was conducted in the ZSL scenario.

When compared to most LLMs, VDM enabled visual models to handle ZSL VDM while maintaining a significantly lower parameter count. This relationship between performance and model efficiency is illustrated in Figure 6.2.

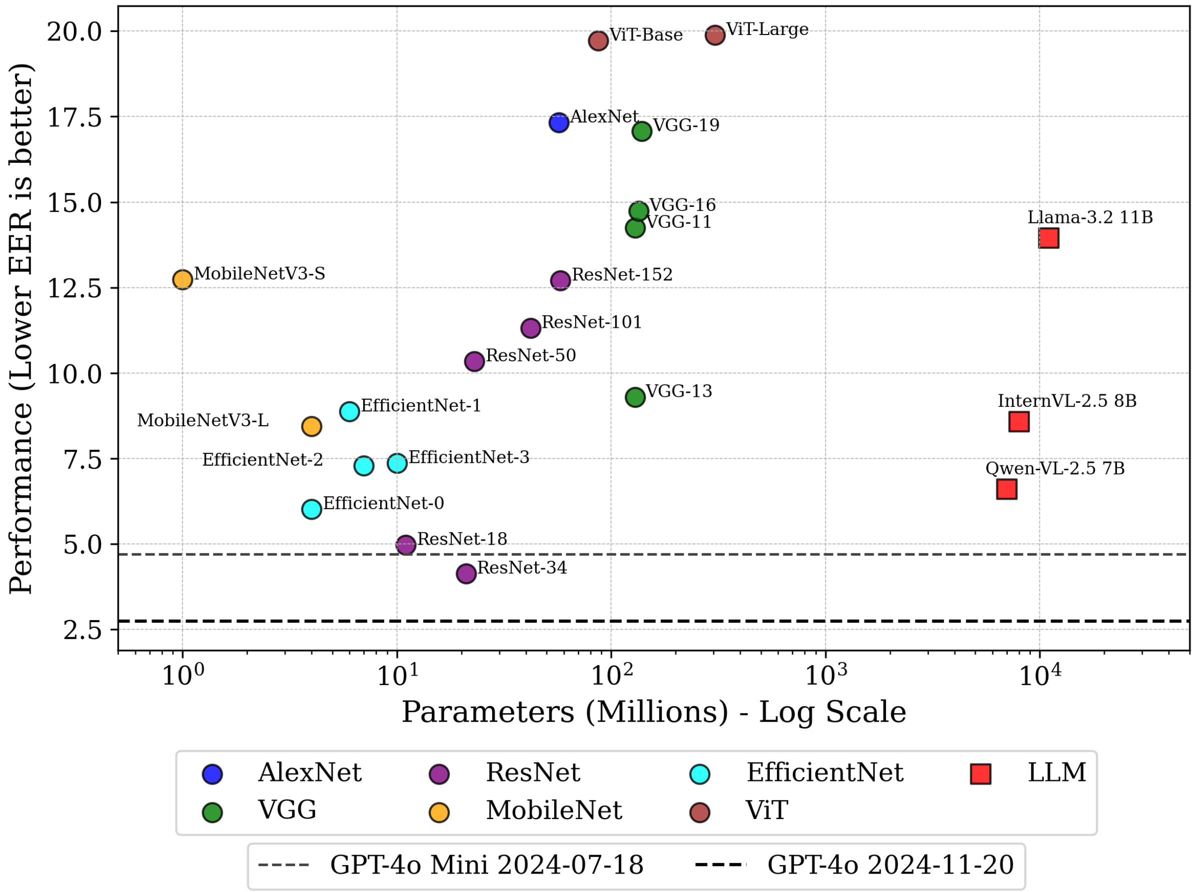


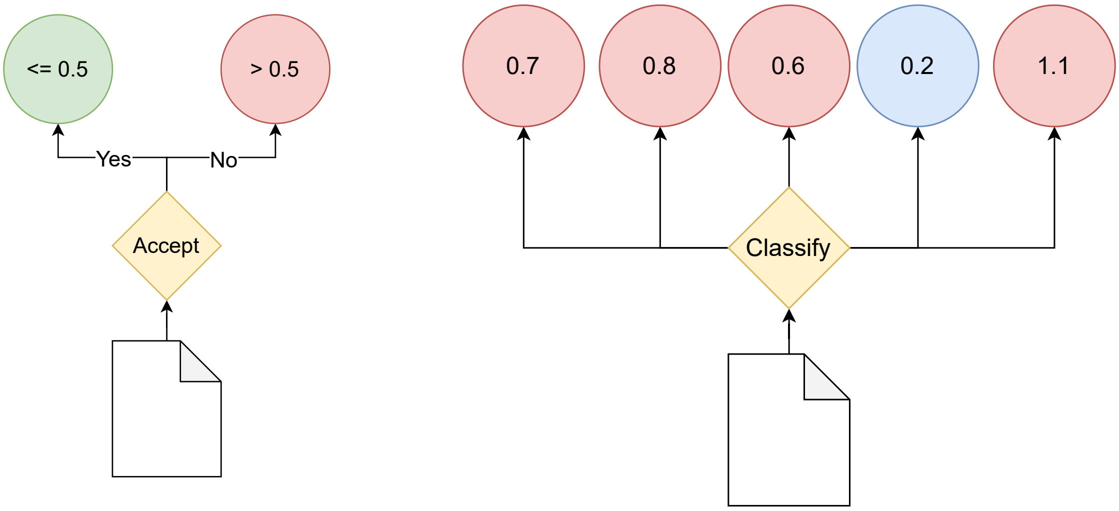
Figure 6.2: Performance vs. Parameters (Zero-Shot Learning scenario). The lines represent GPT-4o and GPT-4o Mini error rates, as their parameter counts were not publicly disclosed.

The results highlight a trade-off between model complexity and effectiveness. Smaller visual backbones, such as ResNet-18 and EfficientNet-0, achieve competitive performance while using significantly fewer parameters than large multimodal models. Notably, ResNet-18 and ResNet-34 exhibit lower error rates than several larger architectures, reinforcing the efficiency of lightweight vision models in document matching tasks.

6.2 Industry Uses

With the trained models shown in this chapter, an industry system can leverage the metric learning nature of the solution to achieve zero-shot DIC in a couple of different ways. First, this model can be used in a verification system, within the same framework used for testing: given a document reference, and with a predefined acceptance threshold, a query document must be accepted if the distance to the reference is smaller than the threshold, and rejected otherwise. As mentioned in 1, this task can be viewed as a binary classification problem, as the system will either accept or reject the document.

Second, the model can be used in an identification system, which becomes similar to a multiclass classification problem: given a document reference for each available class, and with a predefined acceptance threshold, a query document must be assigned to the closest class with respect to the embedding distances, or detected as outside of the label space if no class is within the predefined threshold. Figure 6.3 illustrates the difference between the verification and identification tasks.



(a) Verification task.

(b) Identification task.

Figure 6.3: Comparative illustration between the verification (a) and identification (b) tasks. The verification system approves documents when the distance between the references and the query that is lower than or equal 0.5, and rejects otherwise. The identification system chose the nearest neighbor to classify the query document.

Both tasks can be facilitated by a few design choices. For example, using multiple ground-truth references per class and computing the distance between the query document and the class centroid can improve classification robustness, as it prevents outlier samples from disproportionately representing an entire class. This decision demands more human annotation per class.

Another possible choice is to use a different threshold for each class. While the model is trained to have all classes share the same acceptance threshold, it is natural that some classes in a real-world scenario have less intra-class variance than others. This, of course, increases the maintenance required per class registered in the system, as calculating a unique threshold demands testing. On the other hand, it may be of interest to change the threshold in order to increase the false rejection rate in scenarios where rejecting a correct document is cheaper than accepting an incorrect document, for example.

All these examples share the same need for a database to store the references. It is possible to store the embeddings directly, instead of the document itself, to reduce the number of inferences the model must process. In the verification scenario, the complexity of an inference should increase linearly with the number of references in the class, i.e., the time to calculate the centroid of the class, $O(r)$, where r is the number of references in a given class (the centroid coordinates can be stored to save time). Following the same strategy, in the identification scenario, along with the cost to calculate the centroid, there is also the cost of comparisons per class to find the nearest neighbor, which results in $O(rc)$, where c is the number of classes in the system.

Table 6.1: Comparative performance between different visual backbones and Large Language Models. Following the columns: the architecture name, the architecture edition, if exists, cross-validation over the ZSL scenario, cross-validation over the GZSL scenario, test performance on the ZSL scenario, and test performance over the GZSL scenario. Every value is a mean EER (%) value over the Cross-Validation (CV) folds.

Architecture	Edition	Params	ZSL	GZSL	Test ZSL	Test GZSL
AlexNet		57M	8.92	5.45	17.33	6.31
VGG	11	129M	7.47	5.01	14.24	3.95
	13	129M	7.03	4.79	9.30	3.95
	16	134M	8.29	5.23	14.74	4.82
	19	139M	7.30	4.57	17.08	3.90
ResNet	18	11M	5.03	1.54	4.98	1.51
	34	21M	4.32	2.10	4.13	1.53
	50	23M	6.90	3.39	10.34	2.21
	101	42M	8.20	2.72	11.31	1.98
	152	58M	9.44	3.38	12.70	2.39
MobileNetV3	Small	1M	7.98	5.06	12.74	5.26
	Large	4M	8.16	4.27	8.45	4.43
EfficientNet	0	4M	4.41	2.27	6.02	0.95
	1	6M	3.93	3.54	8.88	2.70
	2	7M	5.73	2.61	7.29	2.14
	3	10M	5.65	3.64	7.37	2.34
ViT	Base	87M	12.43	7.97	19.72	5.19
	Large	305M	13.16	7.57	19.88	5.26
Llama	3.2	11B	–	–	13.95	21.90
InternVL	2.5	8B	–	–	8.58	10.40
Qwen-VL	2.5	7B	–	–	6.61	4.20
GPT 4o mini	2024-07-18	*	–	–	4.70	4.07
GPT 4o	2024-11-20	*	–	–	2.75	1.33

* The parameter count of GPT-4o has not been publicly disclosed.

Chapter 7

Conclusion

This work introduces LA-CDIP, a document image dataset categorized exclusively by the layout information of the document image. This dataset enables VDM research, providing a Zero-Shot alternative to the Document Image Classification problem. The dataset is benchmarked by training siamese networks employing the VDM approach on a diverse catalog of well-established visual backbones and compare their results with popular Large Language Models that require no additional training. Generally LLMs underperform compared to visual models, except with GPT-4o, which achieves a slight performance gain over visual models. Still, it is hundreds of times more expensive per inference, making its advantage less practical.

There are some known limitations of the work and plans to tackle them. First, as mentioned, the complexity of an model architecture is currently an obstacle, as the dataset is relatively small. It is an ongoing work to solve this problem by increase its size, in number of samples per class and number of classes. Increasing the number of document sources by gathering documents from sources other than the RVL-CDIP dataset, also hold value as research, as it increases the generalization of the trained models, and is already mapped as future work. This should allow effectiveness in the bigger models. In the topic of the training pipeline, employing data augmentation techniques should relieve the necessity to expand the dataset, allowing the utilization of greater and more complex models.

7.1 Timeline

This chapter presents the timeline of this master's research. The production of this document happened in the 8th semester. Therefore, everything that came before represents the past: classes, steps of the research and experiments. Everything that comes after is the intended work and next steps, culminating on the thesis defense, in the 10th semester.

Task Description	Trimester									
	1º	2º	3º	4º	5º	6º	7º	8º	9º	10º
Class: Artificial Inteligence 1										
Class: Seminar										
Class: Algorithm Design and Complexity										
Scope Definition										
Literature Review										
Pretrained Image Models Experiments										
Class: Fundamentals of Computer Systems										
Class: Internship										
Framework Construction										
Dataset Labeling										
Architecture Experiments										
Paper Production										
Extra Dataset Labeling										
Paper Presentation										
Qualification										x
Master's Thesis Defense										

Table 7.1: The timeline of the master's research. The "x" represents the current moment in the timeline.

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